

ISHAN BHARDWAJ

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Quantitative finance graduate student specializing in systematic strategy development, backtesting engine design, and live broker execution. Built production-grade trading systems with rigorous IS/OOS validation, fill auditing, and automated signal generation. Targeting systematic trading, quant research, and derivatives roles.

EDUCATION

University of Minnesota — Twin Cities M.S. Financial Mathematics	Expected Dec 2026
KIIT University B.Tech, Computer Science	Jul 2020 – May 2024

SKILLS

Trading & Quant	Systematic strategy development, backtesting engine design, walk-forward IS/OOS validation, IBKR API (ib_insync), ES/NQ/CL futures, options pricing (Black-Scholes, binomial, Monte Carlo), Greeks, implied volatility, delta hedging
ML & Data	Python (pandas, numpy, scikit-learn, XGBoost), SHAP, pyarrow, asyncio, time-series forecasting, feature engineering
Finance	Cointegration analysis (Engle-Granger, Johansen), Kalman filtering, VaR/CVaR, Sharpe/Sortino/Calmar ratios, vol targeting, Kelly criterion

WORK EXPERIENCE

Quantitative Finance Expert (Contractor)	Jul 2026 – Present
Mercor	

- Evaluate AI model outputs on graduate-level quant finance tasks — derivatives pricing, options Greeks, structural credit models (Merton), systematic trading logic — providing authoritative feedback on mathematical accuracy and reasoning quality.
- Design ground-truth benchmarks for stochastic processes, risk model construction (VaR/CVaR/Sharpe), and portfolio optimization; outputs used in training frontier AI systems.

Equity Research Analyst Intern	Jun 2025 – Aug 2025
Atlasio Capital Partners	

- Built predictive financial models across 40+ companies, identifying margin inefficiencies of 300–500 bps; designed automated data workflows reducing inconsistencies by 25%.
- Developed scenario-based pricing and cost sensitivity models across 10+ assumptions; Python pipeline automation improved research turnaround by 20%.

Quantitative Analyst Intern	Jan 2024 – Mar 2024
Mudraksh & McShaw	

- Analyzed 500K+ records identifying statistical trends and cost drivers; built automated pipelines reducing processing time by 20–25% and KPI dashboards tracking 15+ metrics.

Trade Support Analyst Intern	Nov 2023 – Dec 2023
AlgoBulls	

- Validated 1,000+ daily transactions across financial systems with near 100% accuracy; supported audit workflows across 10,000+ transactions.

PROJECTS

CME Systematic Futures Trading System — Top 10%, CME Institute Competition	GitHub
Python 3.12 · pandas · numpy · pyarrow · asyncio · IBKR API (ib_insync) · Custom backtesting engine	

- End-to-end automated futures system across 5 independent strategies on ES/NQ/CL; OOS Sharpe 4.85, IS/OOS ratio 1.34x, profitable in all 5 held-out years (2022–2026). Fill validity audit corrected win rate from 79.9% to honest 53–60%.
- Explored 15 signal types, rejected 11 after OOS failure with documented reasons; 38-check test suite verifying P&L to the cent.

Options Pricing Engine — From Scratch, No QuantLib	GitHub
Python · Black-Scholes · CRR Binomial · Monte Carlo · Brent/Newton IV Solvers · Delta-Hedging Simulation	

- Three independent pricers (BS analytical, CRR binomial w/ American early exercise, MC w/ antithetic variates for European/Asian/barrier) with all 5 Greeks verified to <1e-4 via finite-difference; IV round-trip <1e-6. Delta-hedging simulation validates vol carry formula.

Cointegration & Kalman Filter Pairs Trading	GitHub
Python · Engle-Granger · Johansen Cointegration · Kalman Filter · Walk-Forward IS/OOS Backtest · Hurst Exponent	

- 190 pairs screened across 5 sectors; Kalman filter for dynamic hedge ratio vs rolling OLS. All 8 IS-cointegrated pairs had negative OOS Sharpe — documenting cointegration breakdown across 2021–2026 regime changes as the central research finding.